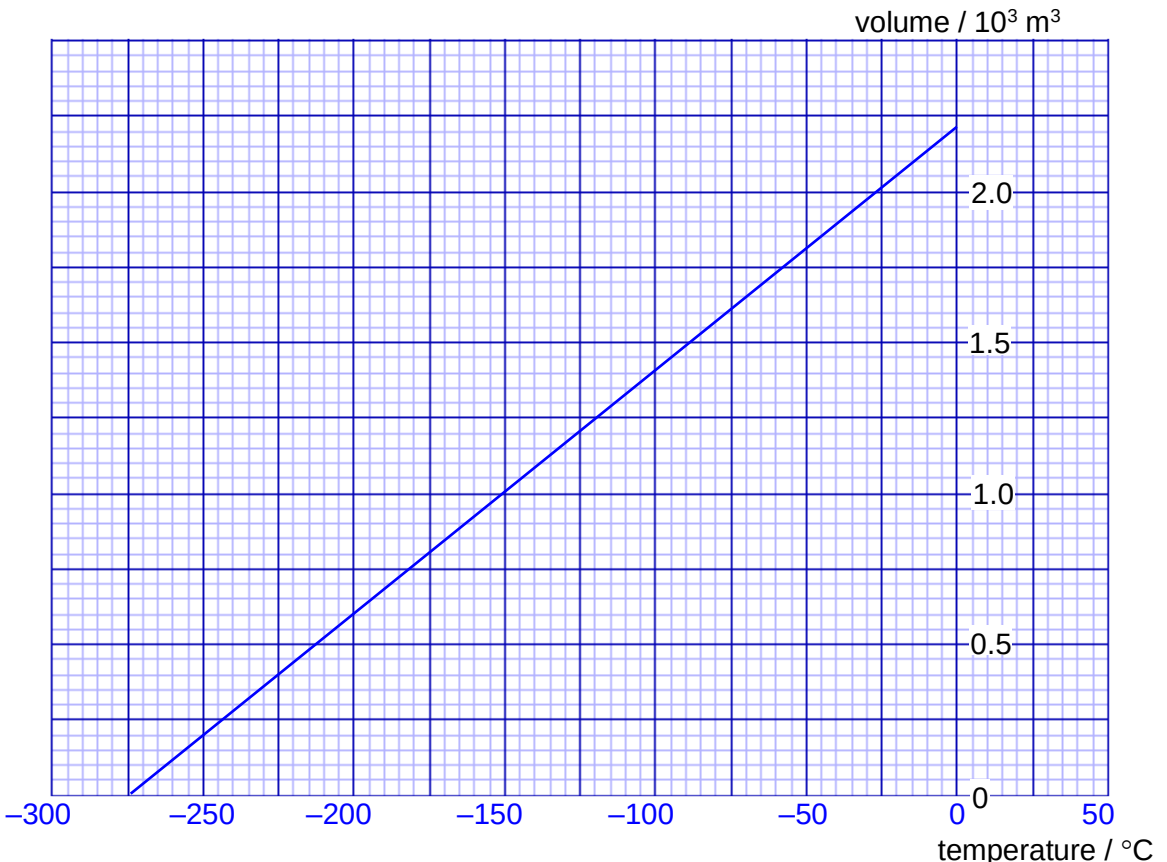


4	A fixed mass of ideal gas at a low temperature is trapped in a container at constant pressure. The gas is then heated and the volume of the container changes so that the pressure stays at $1.00 \times 10^5 \text{ Pa}$. When the gas reaches a temperature of $0.00 \text{ }^\circ\text{C}$ the volume is $2.20 \times 10^{-3} \text{ m}^3$.
	(a) Draw a graph on the axes below to show how the volume of the gas varies with temperature in $^\circ\text{C}$. [2]  graph passes through given point $2.2 \times 10^{-3} \text{ m}^3$ at $0.00 \text{ }^\circ\text{C}$ straight line with positive gradient (straight) line to aim or pass through $-273 \text{ }^\circ\text{C}$ at zero volume
	(b) Calculate the number of moles of gas in the container.
	LO8d $n = P V / R T$ $= 1.00 \times 10^5 \times 2.20 \times 10^{-3} / (8.31 \times 273.15) \square$ $= 0.0969 \text{ (moles)}$
	number of moles = mol [2]
	(c) Calculate the average kinetic energy of a molecule when this gas is at a temperature of $50.0 \text{ }^\circ\text{C}$.
	LO8h mean kinetic energy $= \frac{3}{2} K T$ $= \frac{3}{2} \times 1.38 \times 10^{-23} \times 323.15 \square$ $= 6.69 \times 10^{-21} \text{ (J)}$

		kinetic energy = J	[2]
	(d)	Hence or otherwise, calculate the total internal energy of the gas at a temperature of 50.0 °C.	
		total internal energy = $6.69 \times 10^{-21} \times 0.0969 \times 6.02 \times 10^{23}$ = 390 (J)	
		internal energy = J	[2]
	(e)	By considering the collisions of gas molecules with the walls of the container, explain why the volume of the container must change if the pressure is to remain constant as the temperature increases.	
		As mean square speed of molecules is proportional to temperature, as temperature increases so does change of momentum or change in velocity, hence pressure will increase.	
			
		Increasing the volume compensates for longer time between collisions (rate of collision decreases) as the temperature increases, also as the volume increases the surface area increases which reduces the pressure	
			
			
			[2]
5		A graph of the voltage input to an ideal transformer is shown in Fig. 5.1. The 45:1 step-down	